

Transfer Learning Using AlexNet, VGG16, ResNet50 and EfficientNetB0

Image Classification Assignment

1. Objective

To implement transfer learning using pre-trained AlexNet, VGG16, ResNet50, and EfficientNetB0 models for image classification and compare their performance using appropriate evaluation metrics.

2. Dataset Used

The CIFAR-10 dataset is used for this experiment. CIFAR-10 contains 60,000 32×32 color images divided into 10 classes. The images are resized to 224×224 so that they can be processed by the ImageNet-pretrained models.

Property	Details
Dataset	CIFAR-10
Task	10-class image classification
Classes	Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck
Pre-trained weights	ImageNet
Input size	224 × 224 × 3
Framework	PyTorch + torchvision
Transfer learning	Frozen feature extractor + new 10-class classifier
Practical subset	10,000 training and 2,000 testing images by default

3. Theory: Transfer Learning

Transfer learning is a deep-learning technique in which knowledge learned by a model on a large source dataset is reused for a different target task. Instead of training a CNN from the beginning, a model pre-trained on ImageNet can provide useful low-level and high-level visual features.

In this assignment, the pre-trained feature extraction layers are frozen and the original ImageNet classification layer is replaced with a new classifier containing 10 output classes for CIFAR-10.

4. Models Used

Model	Main Characteristics	Transfer-Learning Modification
AlexNet	Early deep CNN architecture with convolutional and fully connected layers.	Freeze convolutional features and replace final classifier.
VGG16	Deep CNN using stacks of 3×3 convolution filters.	Freeze convolutional features and replace final classifier.
ResNet50	50-layer residual network using skip connections.	Freeze pre-trained layers and replace final fully connected layer.
EfficientNetB0	Efficient architecture balancing depth, width and image	Freeze feature extractor and replace final classifier.

5. Methodology

1. Load the CIFAR-10 dataset.
2. Resize images from 32×32 to 224×224.
3. Normalize images using ImageNet mean and standard deviation.
4. Load ImageNet-pretrained AlexNet, VGG16, ResNet50, and EfficientNetB0.
5. Freeze the pre-trained feature-extraction layers.
6. Replace the original 1,000-class ImageNet classifier with a 10-class CIFAR-10 classifier.
7. Train only the new classification head using the CIFAR-10 training data.
8. Evaluate each model on the same test subset.
9. Calculate accuracy, precision, recall, and F1 score.
10. Generate confusion matrices and compare training time and parameter counts.

6. Preprocessing

Training images are resized to 224×224, randomly horizontally flipped, converted to tensors, and normalized. Test images are resized, converted to tensors, and normalized without random augmentation.

ImageNet normalization uses mean = [0.485, 0.456, 0.406] and standard deviation = [0.229, 0.224, 0.225].

7. Model Training

The models are trained using the Adam optimizer with a learning rate of 0.001 and Cross-Entropy Loss. The default practical configuration uses 3 epochs and a batch size of 64. These settings can be increased when more training time is available.

```
# Example transfer-learning setup
model = models.resnet50(
    weights=models.ResNet50_Weights.DEFAULT
)

# Freeze pre-trained parameters
for param in model.parameters():
    param.requires_grad = False

# Replace ImageNet classifier
model.fc = nn.Linear(
    model.fc.in_features,
    10
)
```

```

criterion = nn.CrossEntropyLoss()

optimizer = optim.Adam(
    [p for p in model.parameters() if p.requires_grad],
    lr=0.001
)

```

8. Evaluation Metrics

Accuracy: The proportion of correctly classified images among all test images.

Precision: Measures how many images predicted as a class are correctly assigned to that class. Weighted precision is used for the multi-class summary.

Recall: Measures how many actual images of the classes are correctly identified. Weighted recall is used.

F1 Score: The harmonic mean of precision and recall. Weighted F1 is reported for the multi-class problem.

Confusion Matrix: Shows the number of correct and incorrect predictions for each class and helps identify classes that are frequently confused.

9. Performance Comparison

The notebook produces a final comparison table containing accuracy, precision, recall, F1 score, training time, total parameters, and trainable parameters for all four architectures.

Model	Accuracy	Precision	Recall	F1 Score
AlexNet	Generated by notebook	Generated by notebook	Generated by notebook	Generated by notebook
VGG16	Generated by notebook	Generated by notebook	Generated by notebook	Generated by notebook
ResNet50	Generated by notebook	Generated by notebook	Generated by notebook	Generated by notebook
EfficientNetB0	Generated by notebook	Generated by notebook	Generated by notebook	Generated by notebook

The numerical values should be filled from the executed notebook. Exact results can vary because of GPU availability, initialization, data ordering, and training settings.

10. Analysis

- AlexNet provides a relatively simple historical CNN architecture and serves as a baseline for transfer learning.
- VGG16 is deeper and has a large number of parameters, which can increase memory and computation requirements.
- ResNet50 uses residual/skip connections, allowing a much deeper network to be trained effectively.
- EfficientNetB0 is designed to achieve a useful balance between accuracy and computational efficiency.
- Because the same dataset, preprocessing, optimizer, learning rate, and epoch count are used, the comparison is more consistent.

- A model should not be judged using accuracy alone; precision, recall, F1 score, confusion matrices, training time, and parameter counts provide a broader analysis.

11. Advantages of Transfer Learning

- Reduces training time compared with training a large CNN from scratch.
- Requires less target-domain data to obtain useful features.
- Makes it practical to use powerful CNN architectures on student-level hardware.
- Pre-trained ImageNet features can provide strong visual representations for many image tasks.

12. Limitations

- ImageNet and CIFAR-10 have different image characteristics and class distributions.
- Freezing all feature-extraction layers may limit adaptation to the target dataset.
- Fine-tuning selected deeper layers may improve results but requires more computation.
- The practical subset and short training schedule are intended for an assignment demonstration rather than production-level benchmarking.

13. Conclusion

Transfer learning was successfully implemented using ImageNet-pretrained AlexNet, VGG16, ResNet50, and EfficientNetB0 models for CIFAR-10 image classification. The original classification heads were replaced with 10-class classifiers while the pre-trained feature extractors were frozen. The models were compared using accuracy, precision, recall, F1 score, confusion matrices, training time, and parameter counts. This experiment demonstrates how pre-trained CNN architectures can be adapted efficiently to a new image-classification task.